

MILESTONE 3:

Milestone 3: Step-by-Step Overview & Definitions

Objective:

Milestone 3 focuses on implementing the Ingredient Intelligence Engine, the Product Recommendation Engine, the Progress Tracking & Photo Pipeline, and building the Interactive Dashboards for both users and dermatologists/consultants.

Step 1: Implement the Ingredient Intelligence Engine

Definition & Purpose

The Ingredient Intelligence Engine evaluates skincare formulas (INCI lists), flags dangerous active clashes, checks for user allergens, and provides educational feedback.

Implementation Steps

Knowledge Base Creation: Seed your database with common active skincare ingredients (Retinoids, AHAs/BHAs, Vitamin C, Niacinamide, Hyaluronic Acid, Ceramides, Peptides) and define their attributes, irritation risks, and incompatible active categories.

Allergy Matching Engine: Build logic to cross-reference user-submitted sensitivity profiles against product ingredient lists to flag direct matches or known aliases.

Chemical Conflict Matrix: Define strict safety rules to detect incompatible ingredient pairings within the same routine step (e.g., combining Retinoids and strong AHAs/BHAs in the same evening step).

Safety Score Endpoint: Create an API endpoint that receives a product's ingredient list and routine time, returning an overall safety score (0–100), status label (Safe, Warning, Unsafe), allergy alerts, and detailed interaction warnings.

Step 2: Implement the Product Recommendation Engine

Definition & Purpose

The Product Recommendation Engine filters, scores, and ranks catalog products to deliver personalized recommendations tailored to skin concerns, skin type, safety profile, and user budget.

Implementation Steps

1. **Catalog Setup:** Organize your product catalog into standard categories (*Face Wash, Moisturizer, Sunscreen, Serum, Toner, Treatment Products, Face Masks*) with mapped active ingredients and price tiers.
2. **Hard-Filter Safety Gate:** Automatically exclude any product from the recommendation pool if it contains user-flagged allergens or unsafe chemical clashes.

3. **Suitability Scoring Algorithm:** Implement a weighted scoring model that evaluates product alignment based on:
 - **Target Concern Match Weight (50%):** How well active ingredients target specific user skin concerns.
 - **Skin Type Fit Weight (35%):** Compatibility with user skin type (e.g., oil-free for oily skin, barrier repair for dry skin).
 - **Rating Weight (15%):** Clinical or user ratings.
4. **Budget Optimization & Alternatives:** Filter results based on maximum price caps, and recommend lower-priced alternative products with similar active profiles if top matches exceed budget limits.
5. **Recommendation API Endpoint:** Build an endpoint that delivers categorized recommendations with match percentages (e.g., *"94% Match"*).

Step 3: Implement Progress Tracking & Cloud Photo Pipeline

Definition & Purpose

This service monitors user consistency, measures skin health improvement over time, and securely manages progress photo uploads for visual tracking.

Implementation Steps

1. **Daily Checklist Logging:** Create a logging workflow in your document database (MongoDB) that records individual check-ins whenever a user completes an AM or PM routine step.
2. **Adherence Math Engine:** Calculate rolling 7-day, 30-day, and 90-day compliance rates based on total completed steps versus total assigned steps.
3. **Cloud Photo Upload Pipeline:**
 - Stream user-uploaded selfie images directly to Cloud Storage (AWS S3 or Azure Blob).
 - Store image metadata (Cloud URL, upload timestamp, skin health score at time of upload, tag like *"Baseline"* or *"Week 4"*) in PostgreSQL.
4. **Analytics API Endpoint:** Expose an analytics endpoint returning historical score timelines, compliance percentages, and progress photo links for charting and comparison.

Step 4: Construct the Frontend Dashboards (React)

Definition & Purpose

The frontend dashboards provide custom interfaces tailored to consumer users and medical professionals/consultants.

Implementation Steps

1. **User Dashboard Construction:**
 - **Top Banner:** Display overall Skin Health Score gauge (0–100) alongside sub-score cards.
 - **Central Checklist:** Interactive AM/PM daily task list with clickable checkboxes that log completion in real time.

- **Analytics Chart:** Render line charts (using Chart.js or Plotly) showing score trajectories alongside adherence rates.
 - **Recommendations Shelf:** Display matching product cards with active ingredient tags and budget flags.
2. **Consultant & Dermatologist Portal Construction:**
- **Patient Roster Panel:** Searchable list displaying assigned clients, their primary concerns, health scores, and compliance metrics.
 - **Patient Inspection View:** Visual timeline panel allowing medical experts to view patient survey details, track adherence, and inspect side-by-side progress photos (*Baseline vs. Current*).
 - **Prescription/Routine Overwrite Form:** Form interface allowing doctors to update user routines, which immediately reflects on the user's daily checklist.

Step 5: Testing & Verification Workflow

Definition & Purpose

Validates system accuracy, safety logic, database persistence, and cross-dashboard synchronization.

Implementation Steps:

1. **Automated Backend Testing:** Execute unit tests to ensure chemical clash detection triggers on unsafe pairings, allergy filters remove unsafe products, and adherence formulas calculate accurately.
2. **End-to-End Walkthrough:**
 - Submit an initial assessment and review generated recommendations.
 - Check off routine tasks and upload a progress photo.
 - Access the dermatologist dashboard, inspect the user's progress photo and compliance stats, and update an evening treatment step.
 - Return to the user dashboard to confirm that the revised routine displays live.

Here are the official outcomes achieved upon completing **Milestone 3 (Product Intelligence, Progress Tracking & User Dashboards)**:

- **Product Recommendation Engine Operational:** Fully functional recommendation pipeline that scores, filters, and ranks products based on user skin profiles, target concerns, and budget limits.
- **Progress Tracking System Functional:** Active calculation of routine compliance/adherence rates alongside cloud storage setup for before/after visual photo progress tracking.
- **Analytics and Recommendation Workflows Completed:** Ingredient safety analysis, allergy detection, active conflict matrix evaluation, and tailored product match workflows fully integrated.

- **Interactive Dashboards Implemented:** Delivered functional web interfaces for consumers (interactive AM/PM checklists, health score gauges, progress charts) and dermatologists/consultants (patient roster, progress photo galleries, routine adjustment forms).

